

SECTION 4.3 SUPPLEMENTAL EXERCISES
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

The following are indicative of the level of difficulty of problems from Section 4.3 that you will find on Exam II (there will not be more than one or two problems from this section).

SECTION 4.3 - EVALUATION OF DEFINITE INTEGRALS

1. Evaluate $\int_{-\infty}^{\infty} \frac{x^2 - 3i}{x^4 + 5} dx$.

2. Evaluate $\int_{-\infty}^{\infty} \frac{\cos(4x) - i \sin(4x)}{6x^2 + 2} dx$.

3. Compute the Cauchy principal value of $\int_{-\infty}^{\infty} \frac{z^2 - 3i}{z^5 + 5} dz$